Online learning of AND under Gaussian neural-network priors

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Abstract

The conjunction of all input bits can force exponentially many online mistakes for a deterministic learner that selects a smallest consistent Boolean circuit. Does the same target defeat exact Bayesian prediction with Gaussian neural-network priors? We prove two contrasting results for fully connected tanh networks at the weak-prior scaling $\gamma = 0$, retaining a global output-margin restriction and exact label conditioning. At each fixed depth, the conjunction has a polynomial negative log function probability, so posterior sampling makes only polynomially many expected mistakes on every input order. Posterior majority and a fixed confidence criterion obey corresponding bounds. These statements hold in the Gaussian-process width limit and at every sufficiently large finite width. In contrast, at depth $O(d)$ the function prior becomes nearly uniform on all Boolean functions: at every sufficiently large width, every ordering of the 2^d inputs incurs at least $(2/5)2^d$ expected sampled mistakes, and every new input fails the strict $2/3$ confidence criterion. This lower bound does not establish majority-prediction errors. An arbitrarily small first-layer variance anisotropy gives a separate strong-prior exponential lower bound, for AND and for a unit-weight one-neuron majority target. Breaking exact variance symmetry is essential to that construction. We also analyze why the minimum-circuit conjunction argument does not identify the minimum-energy neural functions. No counterexample for the equal-variance strong-prior or $\gamma \geq 1$ regimes is proved here.

1 The learning question and the model

The target is the Boolean conjunction $f_\wedge(x) = +1$ only at $x = (1, \ldots, 1)$, on $\mathcal{X} = \{-1, 1\}^d$; it is negative elsewhere. Its minimum binary AND/OR/NOT circuit size is $d - 1$. An online learner sees an input, predicts a label, and receives its true label. Every observation is retained. Inputs may be chosen using the entire past, but before the current private random prediction. A fixed input permutation is an important special case.

For deterministic minimum-circuit selection, a conjunction ladder can force $2^d - d - 2$ mistakes [2]. After all inputs of Hamming weight below k have been labeled negative, and the all-ones input positive, the smallest consistent circuits are k-variable conjunctions. Each has a different remaining weight-k positive input. A query at the selected conjunction's input therefore causes a mistake. Fresh uniform randomization over those circuits changes that conclusion. A neural posterior, however, weights functions by parameter-space probability, not gate count, so neither the deterministic construction nor its uniform-tie remedy transfers without a new argument.

We use $L - 1$ hidden layers of common width N, tanh activations, a linear readout, and one appended constant input. There are no biases in later layers. The independent centered Gaussian weight variances in the one-parameter family are

$$\mathrm{Var}(W^1_{ij}) = N^{1-\gamma}/d, \qquad \mathrm{Var}(W^s_{ij}) = N^{-\gamma}\ (1 < s < L), \qquad \mathrm{Var}(v_i) = N^{-\gamma-1}. \tag{1}$$

The main positive and negative theorems of this note concern $\gamma = 0$. Before observing labels, condition on the label-independent event $|z_w(x)| > a$ for every input; the default is $a = 1/2$. For a function g, write

$$U_g = \{w : g(x)z_w(x) > a \ \forall x\}, \qquad \pi_N(g) = \frac{\mathbb{P}(U_g)}{\mathbb{P}(\bigcup_h U_h)}.$$

Training conditions this fixed function prior on all observed labels. Nothing flattens function ranks, changes mass ratios, or replaces the posterior with an optimizer. A separately labeled final subsection changes the first-layer variances by arbitrarily small unequal factors and proves a strong-prior lower bound for that perturbed prior.

When a bounded-weight teacher complexity is used below, $C_L(f)$ denotes the least hidden width of a depth-L tanh implementation of f whose coefficients all have magnitude at most that width and whose signed outputs exceed $1/2$ everywhere. Thus $N \geq C_L(f)$ is a specific sufficient-width convention, not merely representability by unbounded real weights. The anisotropic majority example satisfies it with $C_2(f) = N = 1$.

A sampled mistake means that a fresh posterior function gives the wrong label. A majority mistake means that the more probable label is wrong, with any stated tie rule. A confidence failure means that the correct label has probability at most $2/3$. Nearly fair predictions can cause many sampling and confidence errors while their slight majority is correct; we keep these claims separate.

The order of limits is explicit throughout: first fix d and a finite L, then let N grow. Choosing $L = L_d = O(d)$ in the lower bound is compatible with this order. The finite-width thresholds are not quantified. In particular, none of these width-limit statements interchanges the strong-prior limit, which instead makes Gaussian variance tend to zero at each fixed width.

2 A polynomial bound for AND at fixed depth

The wide-network Gaussian limit also gives a positive result that restricts where an AND lower bound can occur. Fix $d \geq 1$ and $L \geq 2$, write $m = 2^d$ and $r = L - 1$, and let

$$f_\wedge(x) = +1 \iff x = (1, \ldots, 1), \qquad \Psi(t) = \frac{2}{\pi} \arcsin t,$$
$$K_r(x, y) = \Psi^{\circ r}\left(\frac{1 + x \cdot y}{d + 1}\right).$$

Consider $G \sim \mathcal{N}(0, K_r)$, conditioned before observing labels on $|G(x)| > a$ at every Boolean input. We take $a = 1/2$ in the neural application. This is the $\gamma = 0$ limiting function prior of the actual layer-scaled Gaussian tanh network; the nonpositive-exponent posterior-limit proposition in [1] supplies this limit.

Theorem 1 (Fixed-depth AND target mass)**.** *Put*

$$A = a + \sqrt{2 \log(4m)}, \qquad D_{d,L,a} = A^2(d + 1)(d^2 - d + 1)\left(\frac{\pi}{2}\right)^r. \tag{2}$$

The unconditioned Gaussian probability of the target's full margin event satisfies

$$\mathbb{P}\{f_\wedge(x)G(x) > a \text{ for every } x\} \geq \frac{1}{4} e^{-D_{d,L,a}}. \tag{3}$$

Consequently the globally margin-conditioned function prior has

$$-\log \pi_\infty(f_\wedge) \leq D_{d,L,a} + \log 4. \tag{4}$$

At fixed L and fixed a, the right side is polynomial in d.

Proof. The Taylor coefficients of Ψ are nonnegative, and its linear coefficient is $2/\pi$. Composition preserves nonnegativity and gives linear coefficient $c_r = (2/\pi)^r$. The Schur product theorem therefore implies

$$K_r \succeq c_r \rho, \qquad \rho(x, y) = \frac{1 + x \cdot y}{d + 1}. \tag{5}$$

Indeed, the remaining positive Taylor terms are positive semidefinite entrywise powers of ρ. Thus we may realize the Gaussian vector as

$$G(x) = \sqrt{\frac{c_r}{d+1}} \left(Z_0 + \sum_{i=1}^{d} Z_i x_i \right) + V(x),$$

where $Z_0, \ldots, Z_d$ are independent standard normals and V is an independent Gaussian vector with covariance $K_r - c_r \rho$.

Shift only the Z_i so that the output mean becomes

$$h(x) = A \left(\sum_{i=1}^{d} x_i - d + 1 \right).$$

The sign of h is $f_\wedge$, and $f_\wedge(x) h(x) \geq A$ everywhere. The relative entropy of the shifted joint Gaussian law with respect to the original joint law is

$$\frac{1}{2} \frac{d+1}{c_r} \left(A^2 (d-1)^2 + d A^2 \right) = \frac{1}{2} D_{d,L,a}. \tag{6}$$

This construction also proves the usual squared reproducing-kernel norm bound for h, without needing to invert the full kernel matrix.

Every centered output has variance one. With $T = \sqrt{2 \log(4m)}$, a Gaussian tail bound and a union bound give

$$\mathbb{P}\{\max_x |G(x)| \geq T\} \leq 2m e^{-T^2/2} = \frac{1}{2}.$$

Hence the shifted law gives probability at least $1/2$ to $\mathcal{E} = \{f_\wedge(x) G(x) > a \text{ for every } x\}$. The strict inequalities cause no difficulty: each boundary is a scalar Gaussian equality event and has probability zero.

Let p and $q \geq 1/2$ be the probabilities of $\mathcal{E}$ under the original and shifted laws. Relative entropy decreases under the map that records membership in $\mathcal{E}$, so

$$\frac{1}{2} D_{d,L,a} \geq q \log \frac{q}{p} + (1 - q) \log \frac{1 - q}{1 - p} \geq q \log \frac{1}{p} - \log 2 \geq \frac{1}{2} \log \frac{1}{p} - \log 2.$$

This proves (3). Conditioning on the union of all full-margin function events can only increase the probability of $\mathcal{E}$, proving (4). □

Corollary 2 (Every-order online guarantee, including finite width)**.** *For the limiting prior, let $B = D_{d,L,a} + \log 4$. Against the AND target, on every nonanticipating input-selection strategy with labels revealed after each prediction, ordinary consistency Bayes has*

- *expected total fresh posterior-sampling mistakes at most B;*
- *at most $B/\log 2$ posterior-majority mistakes;*
- *at most $B/\log(3/2)$ rounds on which the correct label has posterior probability at most $2/3$.*

These include infinite streams. For the actual Gaussian tanh networks at $\gamma = 0$, the same statements hold, with B replaced by $D_{d,L,a} + \log 8$, for every sufficiently large finite width N. The width threshold can depend on d, L, a but not on the input history, its order, or its length.

Proof. For any fixed prior π and realizable target f, let p_t be the posterior probability of its correct label at the round-t query. Exact conditioning telescopes to

$$\sum_{t=1}^{T} -\log p_t \le -\log \pi(f).$$

The sampling bound follows from $1 - p_t \le -\log p_t$ and monotone convergence. A majority mistake has $p_t \le 1/2$, while a failure of correct-label probability strictly greater than $2/3$ contributes at least $\log(3/2)$.

For fixed d, L, the unconditioned finite-width output vector converges in distribution to G by the hidden-kernel and Gaussian-readout limit in [1]. The boundary of the full target margin event has limiting probability zero. Its raw probability therefore converges to a value at least $e^{-D_{d,L,a}}/4$, and is at least $e^{-D_{d,L,a}}/8$ for all sufficiently large N. Global margin conditioning again increases the target probability. Applying the same fixed-prior potential identity proves all three bounds simultaneously for every history; no posterior convergence uniform over histories is being assumed. □

Other fixed Gaussian scales and negative exponents. The same argument applies to $G \sim \mathcal{N}(0, \sigma^2 K_r)$ for any fixed $\sigma > 0$. Replace the budget in (2) by

$$D_{d,L,a,\sigma} = \left(\frac{a}{\sigma} + \sqrt{2\log(4m)}\right)^2 (d+1)(d^2 - d + 1)\left(\frac{\pi}{2}\right)^r. \tag{7}$$

Multiplying all layer standard deviations by this fixed σ leaves the saturated hidden-kernel limit unchanged and scales the readout covariance by σ^2, so the finite-width transfer is valid for this model as well. For the $\gamma < 0$ limit from [1], the global margin disappears after output normalization. Set $a = 0$ in the argument; the sign-only limiting prior and all sufficiently large finite widths obey the corresponding bound. The original finite-width margin becomes a vanishing normalized threshold, and the limiting Gaussian assigns zero probability to its boundaries.

These bounds are polynomial in d for each fixed depth and fixed positive scale, but contain the factor $(\pi/2)^{L-1}$. They therefore do not rule out an exponential AND lower bound at depth growing with d. Nor is the finite-width threshold quantified here or uniform as $\sigma \downarrow 0$. This is a width-first statement and does not interchange the strong-prior order of limits.

3 An exponential conjunction lower bound at growing depth

The unchanged Gaussian learner can make exponentially many sampled mistakes on a conjunction when depth grows with dimension. This result uses the weak-prior scaling $\gamma = 0$. It does not address the fixed-depth strong-prior or $\gamma \ge 1$ regimes.

Throughout this section, $d \ge 2$, $\mathcal{X} = \{-1, 1\}^d$, and $m = 2^d$. We use tanh hidden layers, one appended constant input, no later biases, and the independent Gaussian variances

$$\mathrm{Var}(W_{ij}^1) = \frac{N}{d}, \qquad \mathrm{Var}(W_{ij}^s) = 1 \quad (1 < s < L), \qquad \mathrm{Var}(v_i) = \frac{1}{N}. \tag{8}$$

The prior is conditioned on $|z_w(x)| > 1/2$ at every cube input, and training retains every exact label. Predictions use a fresh posterior sample. Define

$$\begin{aligned} k_0(d) &= \left\lceil \frac{\log(d+1)}{\log(4/3)} \right\rceil, \\ k_1(d) &= \left\lceil \frac{\log(50m^2)}{\log(3/2)} \right\rceil, \qquad L_d = 1 + k_0(d) + k_1(d). \end{aligned} \tag{9}$$

In particular, $L_d = O(d)$.

Theorem 3 (Growing-depth Gaussian NNBayes on the big AND)**.** *Let f_{AND} be positive only at $(1, \dots, 1)$ and negative elsewhere. At depth L_d and the variances (8), there exists $N_0(d) \geq d$ such that, for every finite $N \geq N_0(d)$, the following statements hold.*

After every realizable label history, the correct-label posterior probability at each previously unseen input belongs to $[2/5, 3/5]$. Consequently, on every ordering of all m distinct inputs,

$$\mathbb{E}M_m \geq \frac{2}{5}\, 2^d, \qquad \mathbb{P}\{M_m \leq 2^d/5\} \leq e^{-2^d/20}. \tag{10}$$

Every query is also a failure to assign the correct label posterior probability strictly greater than $2/3$. The ordering may be fixed in advance or selected adaptively from previously unseen inputs, before the current private posterior draw.

The target has a same-depth tanh implementation of width d, with every coefficient bounded by d and strict signed output margin greater than $1/2$.

The width threshold is only asserted to exist; no quantitative bound on $N_0(d)$ is proved. The mistake statements concern posterior sampling and posterior confidence. They do not give a posterior-majority mistake lower bound.

Lemma 4 (The fixed-dimensional width limit)**.** *For each fixed d, L, under the raw prior (8), the vector of outputs converges in law to a centered Gaussian vector with covariance*

$$K_L(x, y) = \Psi^{\circ(L-1)}\left(\frac{1 + x \cdot y}{d+1}\right), \qquad \Psi(t) = \frac{2}{\pi}\arcsin(t). \tag{11}$$

Proof. For hidden layer s, write $Q_N^s(x, y) = N^{-1}\sum_{i=1}^N h_i^s(x) h_i^s(y)$. First-layer rows are independent Gaussian fields with covariance $N(1 + x \cdot y)/d$ before applying tanh. As their scale tends to infinity, tanh converges almost surely to the sign of a coupled Gaussian coordinate. The Gaussian sign-correlation identity gives the limiting second moment $\Psi((1 + x \cdot y)/(d+1))$. Boundedness of the summands gives convergence of the empirical second moments in probability, simultaneously over the finitely many input pairs.

Conditionally on a preceding layer, the next preactivation rows are independent Gaussian fields with covariance NQ_N^{s-1}. If the preceding empirical covariance tends to a fixed matrix with diagonal one, the corresponding tanh second moments converge to its entrywise Gaussian sign correlations. This follows by a continuous Gaussian-square-root coupling and dominated convergence; each limiting scalar coordinate is nondegenerate. Conditional variances of the empirical second moments are at most $1/N$. Induction over the fixed number of layers therefore gives $Q_N^{L-1} \to K_L$ in probability. Finally, the conditional readout law is exactly $\mathcal{N}(0, Q_N^{L-1})$. Convergence of its characteristic function proves the claim. □

Although the theorem chooses $L = L_d$, this use of Lemma 4 is legitimate: dimension and its chosen finite depth are fixed before width tends to infinity. No width-limit theorem uniform in increasing depth is required.

Lemma 5 (Covariance close to the identity)**.** *At the depth* (9)*,*

$$\|K_{L_d} - I_m\|_{\mathrm{op}} \le \eta, \qquad \eta = \frac{1}{100m}.$$

Proof. For distinct cube vertices, $|(1 + x \cdot y)/(d+1)| \le 1 - 2/(d+1)$. The map Ψ is odd, increasing, and convex on $[0, 1]$. Since $\Psi(1/2) = 1/3$ and $\Psi(1) = 1$, its chord on $[1/2, 1]$ gives

$$1 - \Psi(t) \ge \frac{4}{3}(1 - t) \qquad (1/2 \le t \le 1).$$

Hence an iterate starting at most $1 - 2/(d+1)$ must fall below or equal to $1/2$ within $k_0(d)$ steps: otherwise its gap from one would be at least $2(4/3)^{k_0(d)}/(d+1) \ge 2$, which is impossible. On $[0, 1/2]$, the chord from zero to $(1/2, 1/3)$ gives $\Psi(t) \le (2/3)t$. The next $k_1(d)$ steps thus make every off-diagonal entry have absolute value at most

$$\frac{1}{2}(2/3)^{k_1(d)} \le \frac{1}{100m^2}.$$

All diagonal entries equal one. The absolute row-sum bound for the operator norm of a symmetric matrix completes the proof. □

Lemma 6 (Uniform comparison of full-margin orthants)**.** *Suppose $G \sim \mathcal{N}(0, K)$ on m coordinates and $\|K - I_m\|_{\mathrm{op}} \le 1/(100m)$. For each sign vector $g \in \{-1, 1\}^m$, put*

$$p_g = \mathbb{P}\{g_i G_i > 1/2 \text{ for all } i\}.$$

Then every p_g is positive and $\max_g p_g / \min_g p_g < 4/3$.

Proof. Let $\eta = 1/(100m)$, $a = 1/2$, and let $\overline{\Phi}(t)$ denote the standard Gaussian upper tail. The eigenvalues of K lie in $[1 - \eta, 1 + \eta]$. Its density is bounded below by

$$(2\pi)^{-m/2}(1 + \eta)^{-m/2} \exp\left(-\frac{\|z\|^2}{2(1 - \eta)}\right)$$

and above by the same expression with the two occurrences of $1 + \eta$ and $1 - \eta$ interchanged. Integrating over any specified full-margin orthant gives

$$\ell \le p_g \le u,$$

where

$$\ell = \left(\frac{1 - \eta}{1 + \eta}\right)^{m/2} \overline{\Phi}\left(\frac{a}{\sqrt{1 - \eta}}\right)^m, \qquad u = \left(\frac{1 + \eta}{1 - \eta}\right)^{m/2} \overline{\Phi}\left(\frac{a}{\sqrt{1 + \eta}}\right)^m.$$

We bound their ratio without discarding the global-margin condition. For $\eta \le 1/2$, $\log((1 + \eta)/(1 - \eta)) \le 3\eta$. Both tail arguments lie in $[0, 1]$. Writing ϕ for the standard Gaussian density, on that interval

$$\frac{\phi(t)}{\overline{\Phi}(t)} \le \frac{\phi(0)}{\int_1^2 \phi(s)\, ds} \le \frac{\phi(0)}{\phi(2)} = e^2 < 8.$$

The difference between the arguments is at most $a\eta(1 - \eta)^{-3/2} \le \sqrt{2}\,\eta$. Integrating the logarithmic derivative of $\overline{\Phi}$ gives

$$\log \frac{u}{\ell} \le (3 + 8\sqrt{2})m\eta < 15m\eta = \frac{3}{20} < \log \frac{4}{3}.$$

The final inequality follows, for example, from $\log(1 + 1/3) \ge 1/4$. □

Proof of Theorem 3. Apply Lemmas 5 and 6 to the Gaussian width limit. For each Boolean function g, define its raw neural probability

$$p_{N,g} = \mathbb{P}\{g(x)z_N(x) > 1/2 \text{ for every } x\}.$$

The limiting Gaussian is nonsingular, so the boundary of each such orthant has probability zero. Lemma 4 therefore gives $p_{N,g} \to p_g > 0$. There are finitely many Boolean functions at fixed dimension. The strict limiting ratio bound implies the existence of $N_0(d)$ such that, for every $N \geq N_0(d)$,

$$\frac{\max_g p_{N,g}}{\min_g p_{N,g}} \leq \frac{3}{2}. \tag{12}$$

Global-margin conditioning divides all these masses by one common normalizer, preserving their ratios.

After s distinct inputs have been labeled, exactly 2^{m-s-1} full Boolean functions extend the history with either specified label at a new input. Equation (12) therefore places the posterior probability of either label in $[2/5, 3/5]$. This argument is uniform over all histories and does not depend on their probability or on the target. Conditional expectation gives the first bound in (10).

For the second bound, let I_t be the sampled mistake indicator. Conditionally on the past and the current previously unseen input,

$$\mathbb{E}[2^{-I_t} \mid \text{past, current input}] = 1 - \frac{1}{2}\mathbb{P}(I_t = 1 \mid \text{past, current input}) \leq \frac{4}{5}.$$

Iterating and applying Markov's inequality gives

$$\mathbb{P}\{M_m \leq m/5\} \leq 2^{m/5}(4/5)^m \leq e^{-m/20},$$

using $\log 2 < 3/4$ and $\log(4/5) \leq -1/5$. The confidence claim follows from $3/5 < 2/3$.

Finally, realize f_{AND} with one active neuron in each hidden layer, padding to width d. The first active neuron computes $\tanh(\sum_j x_j - (d-1))$; each later active neuron computes $\tanh(2h)$ from the previous one; the readout has coefficient one. The first signed activation is at least $\tanh(1) > 1/2$. Since $2\tanh(1) > 1$, subsequent layers preserve a signed activation of magnitude at least $\tanh(1)$. Every coefficient has magnitude at most d for $d \geq 2$, and the desired strict margin follows. □

The theorem applies, in particular, to a fixed lexicographic ordering of the cube. It does not use a circuit tie rule or inspect a sampled hypothesis before choosing its query. The conjunction has $d-1$ Boolean AND gates and only $O(d + L_d)$ nonzero coefficients in the displayed neural implementation. Its exponential mistake count is compatible with fixed-depth polynomial theorems because L_d grows with d, and because (8) lies at $\gamma = 0$.

4 Exact width-one geometry at the start of an AND layer

This section concerns a width-one network with one hidden tanh unit,

$$z_{a,b}(x) = b\tanh\left(a_0 + \sum_{i=1}^{d} a_i x_i\right), \qquad x \in \{-1,1\}^d, \qquad \kappa(a,b) = d\sum_{i=0}^{d} a_i^2 + b^2.$$

The prior is the centered Gaussian with density proportional to $\exp[-\kappa(a,b)/(2\sigma^2)]$, conditioned before training on $|z_{a,b}(x)| > 1/2$ at every cube input. For a Boolean threshold function g, define

its affine unit-margin cost

$$Q(g) = \min\left\{\sum_{i=0}^{d} a_i^2 : g(x)\left(a_0 + \sum_{i=1}^{d} a_i x_i\right) \geq 1 \text{ for every } x\right\}.$$

The minimum exists: the feasible set is a nonempty closed polyhedron and the objective is coercive. Non-threshold functions are not represented by this width-one architecture.

Lemma 7 (Reduction to affine margin cost)**.** *For every represented Boolean function g, its minimum strict-margin neural energy is*

$$E(g) = \Phi_d(Q(g)), \qquad \Phi_d(Q) = \min_{u>0}\left\{dQu^2 + \frac{1}{4}\coth^2 u\right\}.$$

The function Φ_d is strictly increasing on $Q > 0$.

Proof. Replacing (a, b) by $(-a, -b)$ preserves the real output and energy, so we can take $b > 0$. A strict output margin requires $b > 1/2$. Put $u = \operatorname{arctanh}(1/(2b)) > 0$. The constraints imply $g(x)(a_0 + \sum_i a_i x_i) > u$, so $\sum_i a_i^2 \geq u^2 Q(g)$ and $b^2 = \frac{1}{4}\coth^2 u$. Conversely, multiply a unit-margin minimizer for g by $(1+\epsilon)u$ and take $b = \frac{1}{2}\coth u$. This gives strict output margin, with energy tending to the displayed expression as $\epsilon \downarrow 0$. This proves equality of the infima. For $Q > 0$ the objective tends to infinity at both endpoints $u \downarrow 0$ and $u \to \infty$, so its minimum is attained at some $u_Q > 0$. If $Q_2 > Q_1$, then

$$\Phi_d(Q_2) = dQ_1 u_{Q_2}^2 + \tfrac{1}{4}\coth^2 u_{Q_2} + d(Q_2 - Q_1)u_{Q_2}^2 > \Phi_d(Q_1).$$

□

Small-variance function probabilities. For this finite-dimensional Gaussian law, the function event $U_g = \{(a, b) : g(x) z_{a,b}(x) > 1/2 \ \forall x\}$ satisfies

$$\lim_{\sigma \downarrow 0} \sigma^2 \log \mathbb{P}_\sigma(U_g) = -\frac{1}{2}E(g). \tag{13}$$

Indeed, the lower Gaussian large-deviation bound applies to the open set U_g, and the upper bound applies to its closure. Their energy infima coincide: any parameter vector with all signed outputs at least $1/2$ can be approximated from within U_g by multiplying the readout by $1 + \epsilon$. The standard finite-dimensional Gaussian rate function is half the quadratic energy. The argument holds unchanged for every positive definite diagonal quadratic energy. There are only finitely many represented Boolean functions. Therefore, after conditioning on any fixed label history, all strong-prior posterior mass concentrates on its minimum-energy functions; if the minimum is unique, its posterior probability tends to one.

Write $|x|_+ = |\{i : x_i = +1\}|$. Fix $2 \leq k \leq d$ and consider the history that labels 1^d positive and every input with $|x|_+ < k$ negative. These labels agree with the AND target, which is positive only at 1^d. No labels at other inputs are imposed yet.

Lemma 8 (Essential affine coefficients)**.** *In a minimum-squared-norm unit-margin representation of a threshold function, every nonzero input coefficient is essential to that function, and every essential input coefficient has absolute value at least one.*

Proof. If the function is independent of x_i, then, for each fixed assignment of the other coordinates, both signs of x_i have the same label. Averaging their two signed margin inequalities shows that setting $a_i = 0$ preserves every unit-margin constraint. It strictly decreases squared norm if $a_i \neq 0$. If x_i is essential, two neighboring cube points differing only in x_i have opposite labels. Their affine scores have opposite signs and magnitudes at least one; their difference has magnitude $2|a_i|$. Hence $|a_i| \geq 1$. □

Proposition 9 (All minimizers at a complete lower-layer prefix)**.** *Among all width-one Boolean functions consistent with this history, the smallest affine unit-margin cost is*

$$Q_{d,k} = \begin{cases} 2k-1, & d \geq 2k-1, \\ d + (2k-1-d)^2, & d < 2k-1. \end{cases}$$

If $d \geq 2k-1$, the minimizing functions are precisely

$$h_T(x) = +1 \quad \Longleftrightarrow \quad |\{i \in T : x_i = +1\}| \geq k, \qquad |T| \in \{2k-1, 2k-2\}.$$

Their minimizing affine scores have coefficient one on T, zero elsewhere, and bias zero for $|T| = 2k-1$ or bias -1 for $|T| = 2k-2$. If $d < 2k-1$, there is exactly one minimizing function: the symmetric threshold $|x|_+ \geq k$, with coefficients one and bias $d - 2k + 1$. The corresponding minimum neural energy is $\Phi_d(Q_{d,k})$.

Proof. Choose a minimizing function and its minimum-norm unit-margin affine representation. Lemma 8 applies. Let p be the number of positive input coefficients, write their sum as P, and let B be the sum of absolute values of the negative input coefficients. In particular $P \geq p$.

One has $p \geq k$: the input which is positive exactly on the positive coefficient coordinates has affine score at least that at 1^d, hence at least one. If $p < k$, the history requires this input to be negative, a contradiction.

If $p \geq 2k-1$, the positive coefficients alone contribute squared norm at least $p \geq 2k-1$. If $p \leq 2k-2$, average the negative margin constraints over inputs that are positive on exactly $k-1$ of those p coordinates and negative everywhere else. This gives

$$a_0 - P + B + \frac{2(k-1)}{p} P \leq -1.$$

Since $1 - 2(k-1)/p \leq 0$ and $P \geq p$,

$$a_0 \leq \left(1 - \frac{2(k-1)}{p}\right) P - B - 1 \leq p - 2k + 1 - B \leq p - 2k + 1 < 0.$$

Consequently the total squared norm is at least

$$p + (2k-1-p)^2. \tag{*}$$

For integer $p \leq 2k-2$, this is at least $2k-1$, with equality only at $p = 2k-2$. Together with the preceding branch, equality at $2k-1$ forces either $p = 2k-1$ with zero bias or $p = 2k-2$ with bias -1. It also forces every positive coefficient to equal one and excludes all negative coefficients. The displayed functions realize these bounds: all their affine scores are odd integers, and their positive inputs contain at least k positive coordinates.

If $d < 2k-1$, every admissible p lies between k and $d \leq 2k-2$. The expression in (*) strictly decreases as integer p increases in this range. It is therefore minimized only at $p = d$, giving $d + (2k-1-d)^2$. Equality forces all coefficients to equal one and the bias to equal $d - 2k + 1$. This score also takes odd integer values and realizes the required symmetric threshold. The energy conclusion follows from Lemma 7. □

Why the circuit conjunction characterization fails here. A conjunction of k positive literals has

$$Q(\mathrm{AND}_k) = k + (k-1)^2 = k^2 - k + 1.$$

To verify the lower bound, remove irrelevant coefficients, use essentiality to make each remaining positive coefficient at least one, and average the k negative constraints obtained by flipping one of the k selected coordinates. This forces the bias to be at most $-(k-1)$. Unit coefficients and that bias attain equality. For $k \geq 3$ and $d \geq 2k-1$,

$$Q(\mathrm{AND}_k) - (2k-1) = (k-1)(k-2) > 0.$$

Thus every k-conjunction is strictly more expensive than the actual minimizers at this prefix. The globally smallest circuit functions at the analogous circuit prefix are k-conjunctions; the Gaussian neural energy gives a different answer already in this one-unit model.

Exact masses in this particular minimizing class. Equality of minimum energy alone does not imply equal Gaussian function probabilities. Here, however, an additional exact symmetry does imply it. Ordinary input permutations make all functions with a fixed support size equally probable. To connect the two support sizes, take $j \in T$ with $|T| = 2k-1$ and transform the affine parameters by

$$a'_0 = -a_j, \qquad a'_j = a_0, \qquad a'_i = a_i \quad (i \neq j).$$

This signed column swap preserves the Gaussian law and energy. For $x \in \{-1,1\}^d$, set

$$y_j = -x_j, \qquad y_i = x_i x_j \quad (i \neq j).$$

Then $a'_0 + \sum_i a'_i x_i = x_j(a_0 + \sum_i a_i y_i)$, so oddness of tanh gives $z'(x) = x_j z(y)$. The map $x \mapsto y$ is a bijection, and hence the global-margin event is preserved. Applied to majority on T, the transformed sign is

$$x_j \operatorname{sign}\left(\sum_{i \in T} y_i\right) = \operatorname{sign}\left(-1 + \sum_{i \in T \setminus \{j\}} x_i\right),$$

which is precisely the threshold of k positives on the even support $T \setminus \{j\}$. Thus all functions in the minimizing class have exactly equal prior mass at every positive variance. Equal mass persists among whichever of them survive later label conditioning.

By the fixed-width Gaussian rate theorem and finiteness of the Boolean function class, the strong-prior posterior at this prefix converges to uniform sampling from these minimizers. At positive variance it also assigns mass to higher-energy functions; it is not exactly this uniform minimum-class rule. In other histories, equal energies need not be accompanied by the symmetry just proved.

Corollary 10 (The first minimum-energy tier cannot itself cause an exponential phase)**.** *Suppose $d \geq 2k-1$ and start from the complete lower-layer prefix above. Let*

$$M_{d,k} = \binom{d}{2k-1} + \binom{d}{2k-2}.$$

For the width-one strong-prior limiting posterior, the expected number of sampled mistakes before this minimum-energy tier is exhausted is at most $1 + \log M_{d,k} \leq 1 + (d+1)\log 2$. The bound includes the round which exhausts it. It holds under every adaptive nonanticipating continuation. If $d < 2k-1$ and $k < d$, its sole initial minimum is eliminated by the first weight-k negative example, causing one mistake on that example. If $k = d$, the prefix has already revealed the entire cube and no further mistakes are possible.

Proof. While at least one of the original minimizers survives, it still has energy $\Phi_d(Q_{d,k})$, so every strong-prior posterior is uniform on the surviving original minimizers. Eliminating a fraction $q < 1$ of a uniform finite class reduces its logarithmic size by at least q. The final exhaustion costs at most one. Summing these conditional error probabilities and taking expectations proves the claim. The singleton case follows from the unique symmetric threshold's positive prediction on every weight-k input. □

Limits of the conclusion. The proposition characterizes the first energy tier at a complete lower-layer prefix. After that tier is exhausted, higher-energy functions can become optimal; their complete sequence is not characterized here. Thus the corollary is not an upper bound for the whole Hamming layer or for the entire AND-learning process. All statements concern width one, one hidden tanh unit, and global margin conditioning. For a growing AND target, width one need not satisfy the separate theorem convention $N \geq C_L(f)$, because that complexity also bounds the magnitude of every teacher weight. No extension to all admissible widths, and no exponential lower bound for the original Gaussian neural posterior, is asserted by this width-one calculation.

4.1 An arbitrarily small anisotropy changes the width-one answer

The preceding equality of masses uses exact symmetry. The next result changes the prior and is not a lower bound for the canonical isotropic first-layer law. It also retains width one; it does not satisfy the separate $N \geq C_L(f)$ convention for the growing AND target in general. It does show that a small independent-Gaussian perturbation can turn the same concrete neural minimizing class into an exponential mistake path.

Proposition 11 (Near-isotropic independent-Gaussian variant)**.** *Let $k \geq 2$ and $d \geq 8k$. For every $\eta > 0$, there are coefficients $1 \leq \lambda_i \leq 1 + \eta$, $0 \leq i \leq d$, with the following property. Use a width-one prior with independent centered Gaussian weights*

$$\mathrm{Var}(a_i) = \frac{\sigma^2}{d\lambda_i}, \qquad \mathrm{Var}(b) = \sigma^2,$$

and the same all-input output-margin conditioning. Its quadratic energy is

$$\kappa_\lambda(a, b) = d \sum_{i=0}^{d} \lambda_i a_i^2 + b^2.$$

For the AND target, there is a fixed distinct-input order on which the strong-prior posterior makes at least

$$\frac{\binom{d}{k}}{\binom{2k-1}{k}} \geq 2^k$$

posterior-sampling mistakes with probability one. Posterior majority is wrong on the same rounds.

For every fixed $0 < \zeta < 1/2$, one sufficiently small positive variance makes the actual Gaussian posterior's wrong-label probability at least $1 - \zeta$ on every one of these rounds. Its expected sampled mistakes on them are therefore at least $(1 - \zeta)2^k$, and its majority is wrong on every one. In particular, $k = \lfloor d/8 \rfloor$ gives an exponential lower bound for this perturbed width-one model.

Proof. Start with the history of Proposition 9. Let $\mathcal{C}$ be its minimizing class and $Q_0 = 2k - 1$. Every other consistent threshold function has strictly larger isotropic affine cost. There are finitely many threshold functions, so their costs exceed Q_0 by a positive common gap Δ.

For $\epsilon > 0$, put $\lambda_i = 1 + \epsilon v_i$, where the numbers $v_i \in (0, 1]$ have distinct sums over distinct subsets of $\{0, \dots, d\}$; for example $v_i = 2^{-i}$ has this property. For each fixed threshold function h, let

$$Q_h(\epsilon) = \min \left\{ \sum_i (1 + \epsilon v_i) a_i^2 : h(x)(a_0 + \textstyle\sum_{i \geq 1} a_i x_i) \geq 1 \text{ for all } x \right\}.$$

Its isotropic optimizer a_h is unique by strict convexity. The optimizers at $\epsilon \downarrow 0$ converge to a_h: boundedness follows from comparison with a_h, and every limit point is an isotropic optimizer. Comparing the perturbed and unperturbed objectives in both directions consequently gives

$$Q_h(\epsilon) = Q_h(0) + \epsilon \sum_i v_i (a_h)_i^2 + o(\epsilon).$$

For $h \in \mathcal{C}$, the nonzero squared coordinates of a_h are exactly the indicator of its odd support T, or of its even support T together with the bias index 0. These augmented supports are pairwise different. Their derivative sums are therefore pairwise distinct. Since $\mathcal{C}$ is finite, all its perturbed costs are pairwise distinct for sufficiently small positive ϵ.

For every $h \in \mathcal{C}$, $Q_h(\epsilon) \leq (1 + \epsilon) Q_0$. All other initially consistent functions have $Q_h(\epsilon) \geq Q_0 + \Delta$, because $v_i > 0$. Taking also $\epsilon < \Delta / Q_0$ and $\epsilon \leq \eta$ ensures that every member of $\mathcal{C}$ remains cheaper than every outsider. The affine-to-tanh reduction is unchanged, with $Q_h(\epsilon)$ in place of $Q(h)$, so the same ordering holds for neural energies.

Until $\mathcal{C}$ is exhausted, the currently consistent function of minimum energy is therefore a unique member of $\mathcal{C}$. Choose any k-element subset of its support and query the input positive exactly on that subset. The selected function predicts positive; the AND target is negative because $k < d$. The query is new, since a consistent selected function cannot be positive on a previously revealed negative query. This step eliminates exactly those candidate supports containing the queried k-set.

There are $\binom{d}{2k-1}$ candidates with odd support size $2k - 1$. A single k-set query eliminates at most $\binom{d-k}{k-1}$ of these. Exhausting $\mathcal{C}$ therefore requires at least

$$\frac{\binom{d}{2k-1}}{\binom{d-k}{k-1}} = \frac{\binom{d}{k}}{\binom{2k-1}{k}} \geq \frac{(d/k)^k}{4^k} \geq 2^k$$

queries, each a certain mistake in the strong-prior limit. Use a fixed convention for choosing each k-set; the complete query sequence can then be computed before the run and completed to a permutation of the cube. It does not inspect any current random draw.

Finally, at each of these finitely many histories the minimum-energy function is unique with a positive energy gap. The Gaussian rate argument applies to every positive definite diagonal quadratic energy. Thus, as the common variance tends to zero, the posterior mass of that function tends to one. Taking the minimum of finitely many positive variance thresholds makes its wrong-label probability at least $1 - \zeta$ at every selected round simultaneously. The input sequence and labels are fixed, so posterior randomness at earlier rounds does not alter this argument. □

Corollary 12 (An admissible width-one majority target)**.** *Let $d \geq 17$ be odd and put $k = \lfloor d/8 \rfloor$. The conclusion of Proposition 11 also holds for the target*

$$f_{\mathrm{maj}}(x) = \operatorname{sign}\left(\sum_{i=1}^{d} x_i \right).$$

This target has bounded-weight teacher complexity $C_2(f_{\mathrm{maj}}) = 1$, so the learner width $N = 1$ satisfies $N \geq C_2(f_{\mathrm{maj}})$. For every $\eta > 0$, an independent-Gaussian first-layer covariance perturbation with

$1 \le \lambda_i \le 1+\eta$ *therefore admits a fixed input permutation with at least* $2^{\lfloor d/8 \rfloor}$ *certain strong-prior mistakes. At all sufficiently small positive variances, the expected sampled mistakes on those rounds are at least* $(1-\zeta)2^{\lfloor d/8 \rfloor}$ *for any prescribed* $0 < \zeta < 1/2$*, and posterior majority is wrong on all of them.*

Proof. The network $z(x) = \tanh(\sum_i x_i)$ has one hidden unit, zero bias, and all input and readout weights equal to one. Since d is odd, $|\sum_i x_i| \ge 1$ on the cube, so its signed output margin is at least $\tanh(1) > 1/2$. Thus its bounded-weight teacher complexity is one.

The construction in Proposition 11 uses only the positive label at 1^d and negative labels at inputs of Hamming weight at most k. Because $k < d/2$, all these labels agree with f_{maj}. Consequently every posterior and every forced query in that construction is unchanged when the target is f_{maj}. After that finite sequence, reveal any remaining inputs with their majority labels. The same Gaussian prior, energy gaps, and positive variance thresholds prove the stated bounds. □

Scope of the perturbed examples. The parameters of the width-one target $\tanh(\sum_i x_i - d + 1)$ have magnitude $O(d)$ and strict output margin $\tanh(1) > 1/2$. Thus the example has a simple, polynomially bounded neural teacher, but this AND realization does not establish the bounded-weight width-admissibility convention. Corollary 12 removes that width issue for a different simple target. Both examples change the exact coordinate symmetry of the first layer. Neither is an exponential lower bound for the canonical isotropic prior; no extension of the AND lower bound to all admissible larger learner widths is established.

High probability and ordinary iid learning. The finite-variance lower bound can also force all the designated mistakes with probability at least $1-\delta$: if there are T forced rounds, apply the proposition with $\zeta < \delta/T$ and use a union bound. Choosing also $\zeta < 1/3$ makes every designated round a failure of the strict $2/3$ confidence requirement.

The same model retains polynomial iid sample complexity. Its supported Boolean functions are affine threshold functions. The arrangement of $m = 2^d$ homogeneous hyperplanes in the $(d+1)$-dimensional affine coefficient space gives at most $2\sum_{j=0}^{d} \binom{m-1}{j} \le 2(d+1)m^d$ possible strict sign patterns. For iid examples from any fixed input distribution, a union bound therefore implies that every consistent function has error at most ε, except with probability δ, once

$$n \ge \frac{\log(2(d+1)2^{d^2}) + \log(1/\delta)}{\varepsilon}.$$

This controls posterior-sampling test error for every prior on this class. The test-distribution probability of inputs failing strict $2/3$ confidence is at most three times the posterior-averaged error. Thus the anisotropic example separates polynomial iid learning from exponential online mistakes under an adversarially chosen fixed order; it does not merely use a model that was already statistically hard under iid data.

5 What the results do and do not settle

The depth-$O(d)$ lower bound is a counterexample for the actual independent-Gaussian neural prior. The inputs need no adversarial ordering. Its proof controls full-margin orthant probabilities, not merely unconditioned covariance or Gaussian total variation, so the initial support restriction is retained. It concerns a weak prior; it is not a counterexample to the fixed-depth circuit-prior guarantees for sufficiently small variance, above mean field, or mean field.

At fixed depth in the Gaussian-process regime, the explicit AND upper bound rules out an exponential asymptotic lower bound once width is sufficiently large. It pays a factor $(\pi/2)^{L-1}$ and depends on a fixed common Gaussian scale when such a scale is included. The bound is not uniform in a scale tending to zero, and the eventual-width statement is not an all-width quantitative theorem.

For the equal-coordinate-variance strong-prior learner, and for the $\gamma \geq 1$ family, an exponential arbitrary-order AND lower bound remains unproved. The neural phase calculation identifies a failure of the circuit proof, not a positive theorem for all subsequent phases or all widths. Conversely, an elementary upper bound through $-\log \pi_N(f_\wedge)$ becoming large does not prove many mistakes. The anisotropic one-neuron example shows that an arbitrarily small change in coordinate variances can give exponential mistakes even for a unit-weight, width-one majority teacher, while retaining polynomial iid sample complexity. The variance perturbation is fixed before the common prior scale is made sufficiently small. This does not justify interchanging those two limits or asserting the result at exactly equal coordinate variances.

The structured-input guarantees of [3] remain valid and do not cover arbitrary input sequences.

Verification. The new results have conventional proofs and independent AI-agent mathematical reviews. They are not Lean formalized. Earlier Lean coverage of the Gaussian width limits does not automatically verify these new consequences. The accompanying verification record specifies the checked statements, source hashes, and PDF build checks.

References

[1] ChatGPT-6 Astra, Kaarel Hänni, and Dmitry Vaintrob. *Exponential parity sample complexity below the mean-field scaling.* https://github.com/kaarelh/AI-works/tree/main/neural-network-generalization/main-results/subcritical-exponential.

[2] ChatGPT-6 Astra, Claude, Kaarel Hänni, and Dmitry Vaintrob. *Exponential mistakes and polynomial guarantees for circuit Occam learning.* https://github.com/kaarelh/AI-works/tree/main/neural-network-generalization/works-in-progress/online-mistakes/occam-distribution-shift.

[3] ChatGPT-6 Astra, Claude, Kaarel Hänni, and Dmitry Vaintrob. *Online mistake bounds for Gaussian neural networks and priors preserving their function rankings.* https://github.com/kaarelh/AI-works/tree/main/neural-network-generalization/works-in-progress/online-mistakes/neural-function-priors.